

5.2.14

EE25BTECH11011 - Navya Banavathu

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Question

Solve the following system of linear equations.

$$x - 3y - 3 = 0 \quad (1)$$

$$3x - 9y - 2 = 0 \quad (2)$$

Solution

The given equations of lines are,

$$\begin{pmatrix} 1 & -3 \end{pmatrix} \mathbf{x} = 3 \quad (3)$$

$$\begin{pmatrix} 3 & -9 \end{pmatrix} \mathbf{x} = 2 \quad (4)$$

In matrix form,

$$\begin{pmatrix} 1 & -3 \\ 3 & -9 \end{pmatrix} \mathbf{x} = \begin{pmatrix} 3 \\ 2 \end{pmatrix} \quad (5)$$

Solution

So the augmented matrix is

$$\left(\begin{array}{cc|c} 1 & -3 & 3 \\ 3 & -9 & 2 \end{array} \right) \quad (6)$$

Performing row operations:

$$R_2 \rightarrow R_2 - 3R_1 \implies \left(\begin{array}{cc|c} 1 & -3 & 3 \\ 0 & 0 & -7 \end{array} \right) \quad (7)$$

This corresponds to the equation

$$0x + 0y = -7, \quad (8)$$

which is a contradiction.

Solution

Hence, the system has **no solution**. The two equations can be written as

$$x - 3y = 3 \quad (9)$$

$$x - 3y = \frac{2}{3} \quad (10)$$

which represent two parallel lines with the same slope but different intercepts.

Therefore, The given system of equations is **inconsistent**, and the two lines are parallel.

Code (C)

Part 1

```
#include <stdio.h>

int solve(int a1, int b1, int c1, int a2, int b2, int c2, float*
x, float* y) {
    int det = a1 * b2 - a2 * b1;

    if (det == 0) {
        if (a1 * c2 == a2 * c1 && b1 * c2 == b2 * c1)
            return 2; // coincident
        else
            return 1; // parallel
    } else {
        *x = (float)(c1 * b2 - c2 * b1) / det;
        *y = (float)(a1 * c2 - a2 * c1) / det;
        return 0; // unique solution
    }
}
```

Python Code: Import and Define points

Part 1

```
import ctypes
import numpy as np
import matplotlib.pyplot as plt
import os

# Path to DLL
dll_path = os.path.join(os.getcwd(), "code7.dll")
lib = ctypes.CDLL(dll_path)

# Define argument and return types
lib.solve.argtypes = [ctypes.c_int]*6 + [ctypes.POINTER(ctypes.c_float), ctypes.POINTER(ctypes.c_float)]
lib.solve.restype = ctypes.c_int
```

Part 2

```
# Output variables
x = ctypes.c_float()
y = ctypes.c_float()

# Solve system via DLL
status = lib.solve(1, -3, 3, 3, -9, 2, ctypes.byref(x), ctypes.
    byref(y))

if status == 0:
    print(f"Unique solution: x={x.value:.2f}, y={y.value:.2f}")
elif status == 1:
    print("No solution (parallel lines).")
else:
    print("Infinite solutions (coincident lines).")
```


Part 3

```
# Plot the lines
x_vals = np.linspace(-5, 5, 400)
y1 = (3 - 1*x_vals)/-3
y2 = (2 - 3*x_vals)/-9

plt.figure(figsize=(6,6))
plt.plot(x_vals, y1, label="x - 3y = 3", color="blue")
plt.plot(x_vals, y2, label="3x - 9y = 2", color="red", linestyle=
    "--")
```

Part 4

```
plt.xlabel("x")
plt.ylabel("y")
plt.title("Graph of Two Lines")
plt.legend()
plt.grid(True)
plt.axhline(0, color="black", linewidth=0.5)
plt.axvline(0, color="black", linewidth=0.5)

# Save figure
plt.savefig("fig7.png", dpi=150)
plt.show()
print("Graph saved as fig7.png")
```

Graphical Representation

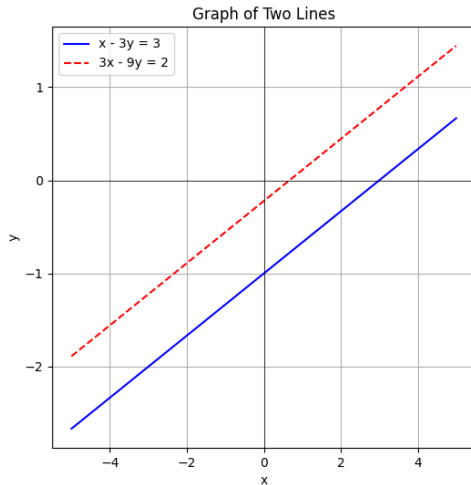


Figure: Parallel lines